

Problem 5A: Calculate the fugacity of water at $\{T = 250^{\circ}\text{C}$ and $P = 70\text{ bar}\}$ with following methods:

1. Steam Tables.
2. Assuming the water follows the Peng-Robinson Equation of State both in the gas and liquid phases.
3. Assuming the water follows the Peng-Robinson Equation of State only in the gas phase.

Note: For methods 2 and 3 you are not allowed to use the steam tables.

$$T = 250^{\circ}\text{C}$$

$$P = 70\text{ Bar}$$

[We can find the Psat and Tsat given T and P]

$$T_{\text{Sat}} = 293.07^{\circ}\text{C}$$

$$P_{\text{Sat}} = 37.07\text{ Bar}$$

[Start with what Gibbs' Free energy is equal to]

$$G = H - TS$$

[Get values for H and S]

$$H = 19.560 \frac{\text{KJ}}{\text{Mol}}$$

$$S = 50.195 \frac{\text{J}}{\text{Mol} * \text{K}}$$

[Plug into the equation at T = 250]

$$G = 19.560 \frac{\text{KJ}}{\text{Mol}} - (250 + 273.15) \left(50.195 \frac{\text{J}}{\text{Mol} * \text{K}} \right) \left(\frac{1\text{kJ}}{1000\text{J}} \right)$$

$$G = -6.699 \frac{\text{KJ}}{\text{Mol}}$$

[Relate Gibbs' Free energy and Fugacity together]

$$dG = RT \ln(F)$$

$$G_2 - G_1 = RT \ln \left(\frac{F_2}{F_1} \right)$$

[We got one G already, but we need to find a second one, we can find one for an Ideal gas as F is P in an ideal gas case]

$$T = 250^{\circ} \text{C}$$

$$P = .1 \text{ Bar}$$

[Get H and S ideal]

$$H = 53.640 \frac{\text{KJ}}{\text{Mol}}$$

$$S = 163.97 \frac{\text{J}}{\text{Mol} * \text{K}}$$

[Do the same calculation as above to get a second G]

$$G = -32.14 \frac{\text{KJ}}{\text{Mol}}$$

[Now we have both Gibbs and one of the fugacity as an ideal fugacity is just pressure]

$$G_2 - G_1 = RT \ln \left(\frac{F_2}{P} \right)$$

$$-6.699 + 32.14 = \left(.08314 \frac{\text{KJ}}{\text{Mol} * \text{K}} \right) (523.15) \ln \left(\frac{f}{.1} \right)$$

[solve for f]

$$\mathbf{f = 34.699 \text{ bar at } T = 250^{\circ}\text{C and at } P = 70 \text{ Bar}}$$

2)

[Pressure and Temperature is the same]

$$T = 250^{\circ} \text{C}$$

$$P = 70 \text{ Bar}$$

[Calculate the Z of both the Liquid and the Vapor]

$$Z_v = .6988$$

$$Z_l = .0449$$

[Then calculate the fugacity coefficient as we have the pressure, temperature, and the compressibility factor]

$$\phi_g = .7716$$

$$\phi_l = .5076$$

[Smaller fugacity constant is only valid as it's the stable phase the substance will take at the pressure and temperature]

$$\phi_l * P = F_l$$

$$.5076 * 70 =$$

$$\mathbf{35.532 \text{ Bar}}$$

3)

[Calculate Psat]

$$P_{sat} = 37.0785 \text{ bar}$$

[we use Psat as a starting guess for EosPsat as we are using Peng-Robinson Equation of State]

$$EOS_{Psat} = 40.1192 \text{ Bar}$$

If we assume that the Peng-Robinson EOS is only valid for GASES then we'll have to use the pointing method, but we'll have to calculate the density first. So, we can use the Racket equation

$$\rho = \frac{RT_c}{P_c} Z^{1+(1-T_r)^{\frac{2}{7}}}$$

$$\rho = 44.87 \text{ Mol/L}$$

$$\frac{1}{\rho} = V = .0223 \frac{\text{L}}{\text{Mol}}$$

The Racket equation only works at the critical temperature but because the liquid is incompressible below the critical temperature. As a result, the density doesn't change, so the specific volume is constant. Then use the Pointing equation

$$f = f_L^{sat} * e^{(V_l/RT(P-P^{Sat}))}$$

$$f = .8823$$

[Multiply the f by the Psat]

$$.8823 * 40.1192 =$$

$$\mathbf{35.3961 \text{ Bar}}$$

All my answer to the 3 parts are very close to each other, so I can say these answers are correct with reasonably high confidence

Problem 5B: Pure propane is maintained at $\{T = 50^\circ\text{C} \text{ and } P = 5\text{ bar}\}$

1. Verify whether propane is in the liquid or gas phase or whether there are two phases present. Assume propane follows the van der Waals equation of state
2. Calculate the numerical value of the derivative $\left(\frac{\partial T}{\partial P}\right)_s$ assuming propane is ideal gas
3. Calculate the numerical value of the derivative $\left(\frac{\partial T}{\partial P}\right)_s$ by using first order numerical differencing and assuming that propane follows the van der Waals equation of state
4. Develop an expression for the derivative $\left(\frac{\partial T}{\partial P}\right)_s$ using the van der Waals equation of state and calculate its numerical value.
5. Prepare a table that shows the results of the three methods.

Note: First order difference formula for the approximation of the first order derivative of a continuous function is given by:

$$\frac{df}{dx} \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

1)

[Start with the pressure and temperature]

$$T = 50^\circ\text{C}$$

$$P = 5\text{ Bar}$$

[Plug in these values into Psat and Tsat]

$$T_{\text{Sat}} = 2.74^\circ\text{C}$$

$$P_{\text{Sat}} = 15.76\text{ Bar}$$

[Using this EosPsat and EosTsat]

$$E_{\text{osTsat}} = -38.97^\circ\text{C}$$

$$E_{\text{osPSat}} = 24.188\text{ Bar}$$

Because our temperature is above the Tsat, **Propane is a gas**

2)

[We need to solve for $\left(\frac{\partial T}{\partial P}\right)_S$, from our first homework we were given]

$$dS = C_p \frac{dT}{T} - R \frac{dP}{P}$$

[Entropy is constant so dS must be zero, solve for $\left(\frac{\partial T}{\partial P}\right)_S$]

$$C_p \frac{dT}{T} - R \frac{dP}{P} = 0$$

$$C_p \frac{dT}{T} = R \frac{dP}{P}$$

$$dT = \frac{RT}{C_p} \frac{dP}{P}$$

$$\left(\frac{\partial T}{\partial P}\right)_S = \frac{RT}{C_p P}$$

[Using ZProp, I can calculate the Cp]

$$C_p = 80.5294 \frac{KJ}{KMol * K}$$

$$R = .0821 \frac{L * Bar}{Mol * K}$$

$$T = 323.15 K$$

$$P = 5 Bar$$

[Plug in values]

$$\left(\frac{\partial T}{\partial P}\right)_S = 6.59 \frac{K}{Bar}$$

3)

[Start with P and T]

$$T = 50 ^\circ C$$

$$P = 5 Bar$$

[Calculate the compressibility factor]

$$Z = .94954$$

[Finally calculate the entropy]

$$S = -277.447 \frac{J}{K}$$

We can change the pressure (dP) by a very small amount, as result entropy will also change. But we already know that entropy must be constant. So the temperature must all change to counter act this. So goal seek the two entropies equal to each other by changing the temperature.

[Final results]

$$T_1 = 50 \text{ } ^\circ \text{C} \quad T_2 = 50.07165 \text{ } ^\circ \text{C}$$

$$P_1 = 5 \text{ Bar} \quad P_2 = 5.01 \text{ Bar}$$

[Because entropy is constant, the $\left(\frac{\partial T}{\partial P}\right)_S$ simplifies]

$$\left(\frac{\partial T}{\partial P}\right)_S = \frac{T_2 - T_1}{P_2 - P_1}$$

[Plug in values]

$$\frac{50.07165 - 50}{5.01 - 5} =$$

[Simplify]

$$\mathbf{7.1650 \text{ K/Bar}}$$

4)

[Start with Van Der Waals entropy equation from class]

$$dS = \frac{C_p}{T} dT - \beta V dP$$

[Under constant Entropy dS becomes zero]

$$0 = \frac{C_p}{T} dT - \beta V dP$$

$$\frac{C_p}{T} dT = \beta V dP$$

$$\left(\frac{\partial T}{\partial P}\right)_S = \frac{\beta V T}{C_p}$$

[Get vales using 301 Calculator]

$$\beta = ,0035$$

$$C_p = 80.5294 \frac{KJ}{KMol * K}$$

$$V = 5.1062 \frac{L}{Mol}$$

$$T = 323.15 K$$

[Plug in values]

$$.0718 \frac{m^3 * K}{KJ}$$

[Unit conversions]

$$KJ = Km * N$$

$$\frac{m^2 * K}{KN}$$

$$Pa = \frac{N}{m^2}$$

$$\frac{K}{KPa} * \left(\frac{100 KPa}{Bar} \right)$$

$$7.18 \frac{K}{Bar}$$

5)

Compare the results

Ideal	Analytical	Van Der Waals
6.5900	7.1650	7.1800

The analytical and Van Der Waals were almost the same. This is to be expected as they both use the Van Der Waal Equation of State. The ideal was clearly different from the rest, but close enough as it is a less accurate equation of state.